

Predictive Analysis of Loan Defaults Using Penalized Logistic Regression, Random Forest, and Gradient Boosting Models

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1 Objective of analysis

The objective of this project is to analyze a loan dataset, with a primary focus on understanding the determinants of loan status (default or no default). The loan status indicates whether a loan is in default or not in default. This analysis aims to investigate the effects of various control variables on the loan status using both parametric and non-parametric approaches.

By employing these methods, we aim to identify significant factors that influence loan default and assess the robustness of our findings across different modeling techniques. Throughout this analysis, we assume exogeneity of the control variables, implying that the predictors are not correlated with the error terms. Consider there is no available information of the randomness level of this dataset (if it's a sample) or if this is the full population to analyse.

The subsequent sections will present a detailed exploration of the dataset, followed by the application of statistical methods to analyze the relationship between control variables and loan status.

2 Data description

Table 1 presents the available information about costumer loans, including costumer demographics, loan details, and default status. Analysis performed in future sections examine how independent variables affect the likelihood of a loan defaulting. In fact, the dataset contains 5,368 units at an individual (costumer) level. Also, the outcome variable is 1 when loan status is “DEFAULT” and 0 when “NO DEFAULT”.

Table 1: Summary of Variables for Data Analysis

Variable Name	Description
Independent Variables	
customer_age	Age of the customer
customer_income	Annual income of the customer
home_ownership	Home ownership status (e.g., RENT, OWN, MORTGAGE)
employment_duration	Duration of employment in months
loan_intent	Purpose of the loan (e.g., PERSONAL, EDUCATION, MEDICAL, VENTURE)
loan_grade	Grade assigned to the loan (e.g., A, B, C, etc.)
loan_amnt	Loan amount requested
loan_int_rate	Interest rate of the loan
term_years	Loan term in years
historical_default	Indicates if the customer has a history of default (Y/N)
cred_hist_length	Length of the customer's credit history in years
Outcome Variable	
Current_loan_status	Current status of the loan (DEFAULT, NO DEFAULT)

3 Preliminary analysis

Before applying modeling techniques, a preliminary analysis was conducted to understand the dataset and the relationships between its variables. This initial examination aims to identify potential patterns, trends, and anomalies that may influence the modeling process.

The outcome variable available in the data is unfortunately, not proportional, as seen in Figure 1. In fact, 78.68% of costumers have their loan in a no default status while the 21.31% are currently in a default status. This non proportional data is still useful since exogeneity is assumed. Although a 50/50 situation would be desirable.

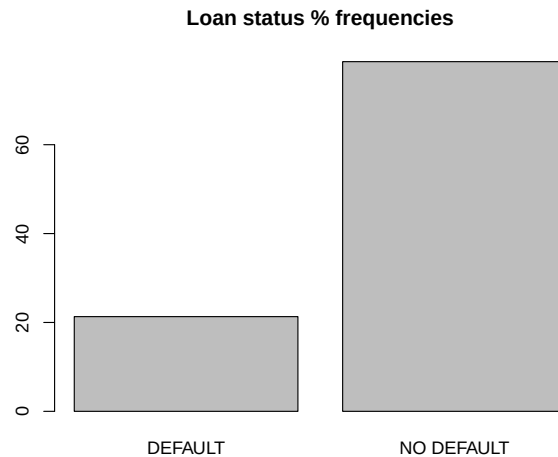


Figure 1: Outcome proportion plot (upper part is NO DEFAULT and lower part is DEFAULT)

Figure 2 plots continuous control variables against the outcome variable loan status. These are: income, employment duration, loan amount, interest rate, loan term and credit history length. Consider the upper part of each plot is corresponds to no default and the lower part is default status. Table 2, 3, 4 and 5 describe proportionally the outcome within each categorical variable.

It is possible to highlight first impressions. For the case of continuous variables, all values in credit history seem to distribute potentially the same for the possible values of the outcome variable. This suggest a weak relation between the mentioned continuous variable and the loan status (default or no default). Although this is not true for the other variables. Specifically variables such as income, employment duration, loan amount, interest rate and loan term tend to distribute differently for costumers with default and no default loan status, suggesting, then, that this variables have an effect on the probability of people incurring in default.

For the case of categorical variables, default and no default status tend to really change for different values of loan grade and historical default. Suggesting, then, strong relation on default status and consumer's historical background in loans. This is also true for the grade of the loan. Nevertheless, this strong relation with outcome variable seem not to be true for categorical variables such as home

ownership and loan intent.

In addition, variables such as *Income* and *Employment Duration* contain outliers that may affect models implemented in further sections.

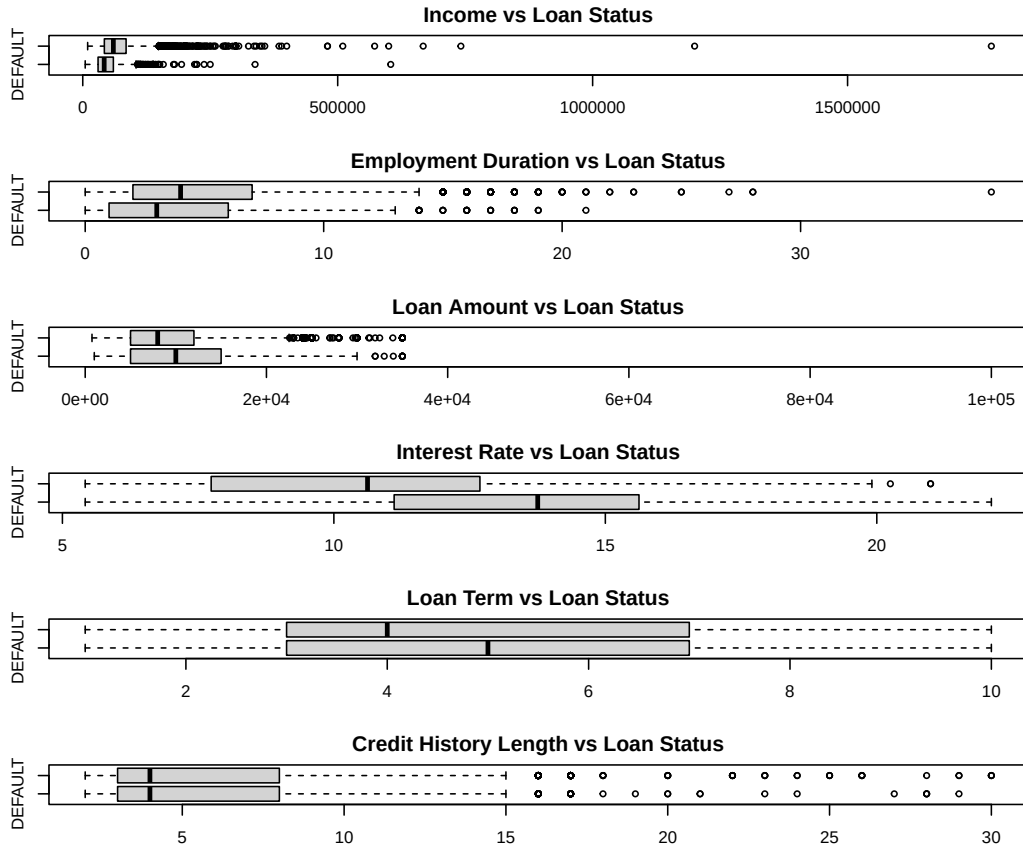


Figure 2: Continuous control variables vs outcome

Table 2: Proportions of Home Ownership and Current Loan Status

Home Ownership	DEFAULT	NO DEFAULT
MORTGAGE	0.1122	0.8878
OWN	0.0466	0.9534
RENT	0.3163	0.6837

Table 3: Proportions of Loan Intent and Current Loan Status

Loan Intent	DEFAULT	NO DEFAULT
DEBTCONSOLIDATION	0.3355	0.6645
EDUCATION	0.1559	0.8441
HOMEIMPROVEMENT	0.2548	0.7452
MEDICAL	0.2416	0.7584
PERSONAL	0.1735	0.8265
VENTURE	0.1595	0.8405

Table 4: Proportions of Loan Grade and Current Loan Status

Loan Grade	DEFAULT	NO DEFAULT
A	0.0995	0.9005
B	0.1904	0.8096
C	0.3465	0.6535
D	0.6190	0.3810
E	0.7761	0.2239

Table 5: Proportions of Historical Default and Current Loan Status

Historical Default	DEFAULT	NO DEFAULT
N	0.7994	0.2006
Unknown	0.0000	1.0000
Y	0.3690	0.6310

It is important to remark that these are just primary highlights from observing data. It is helpful to get an overall view of what potentially affects or not the probability of default. However, next section performs parametric and non parametric to draw robust conclusions.

4 Modeling

4.1 Logistic regression

Due to the binary form of the outcome variable, it is necessary to perform a logistic regression to estimate the form of $\hat{f}(x_i)$ that correctly models $y_i = f(x_i) + \epsilon_i$. There was no model selection performed in this

approach, which means all control variables available are included. Even though, a following parametric model approach will consider model selection, this primary estimation reveal a benchmark for more complex models. Therefore the model performed was:

$$y_i = \beta^T \mathbf{X} + \epsilon_i \quad (1)$$

Equation (1) reveals the first parametric approach performed, where $\mathbf{X}$ includes all covariates from Table 1. Also, no interactions are included since the available variables seem insignificant in any form of interaction. Estimations of this model are presented in the next section since equation (1) is usually a very simple model, thus estimations can be easily improved with more complex model. However, it's still useful as a benchmark.

4.2 Penalized regression: Lasso

A more complex parametric approach is the penalized regression. The purpose of doing so is to reduce error in prediction by specifying a more accurate model for the outcome of interest. This means including and excluding covariates until a final model is reached. This model should include the actual relevant variables that explain the outcome. In our case, we are interested in control variables that actually have an important effect on costumers leading to default in their loans. The model estimated for this is equation (1), although a hyperparameter λ is included to perform penalization. This is, generally speaking a Loss Function such as:

$$AvarageLoss(\beta) + \lambda \sum_{j=1}^{np} |\beta_j|.$$

Also, a k-fold cross validation was jointly performed for the correct selection of λ . Thus, the risk of over fitting is addressed. The hyper parameter λ was set to 0.003697864, since it best performs the penalized model. In other words, the decision of the λ was based on choosing λ_i giving the smallest $\overline{MSE}_i$ among those that are greater or equal to $MSE(\lambda_{min}) + SD(\lambda_{min})$.

Table 6 and 7 show summaries on the Lasso-penalized regressed model and the non penalized general regression. These shown important changes. Note that two of the control variables are now excluded, due to their irrelevance. Therefore no estimation is presented for the *loanitentHOMEIMPROVEMENT* and the *loanamnt* controls. This suggest the irrelevance of these controls due to model specification performed by Lasso penalization. Also, estimates are pulled toward 0.

It's important to have in mind that the main goal of performing this more complex penalized-parametric approach is to improve predictive errors (reducing them). Table 7 reveals a very similar predictive classification performance. Although, ironically, AUROC and Brier are higher for the Non Penalized Full Model. This suggest that prediction in test data is slightly better in the very first model performed without penalization.

Table 6: Comparison of Coefficient Estimates from Logistic LASSO and Logistic Non Penalized Regression

Variable	LASSO Coefficient	Non Penalized Coefficient
(Intercept)	3.8333	5.6762
customer_age	−0.0662	−0.0848
customer_income	0.0000	0.0000
home_ownershipOWN	−1.0707	−1.6613
home_ownershipRENT	1.0577	1.2932
employment_duration	−0.0205	−0.0470
loan_intentEDUCATION	−0.6439	−1.8013
loan_intentHOMEIMPROVEMENT		−0.8652
loan_intentMEDICAL	−0.0987	−1.1894
loan_intentPERSONAL	−0.6847	−1.9476
loan_intentVENTURE	−0.3552	−1.5291
loan_gradeB	−0.5376	−0.5234
loan_gradeC	0.2167	0.7646
loan_gradeD	1.7699	2.6703
loan_gradeE	1.6726	2.4172
loan_amnt		0.0001
loan_int_rate	0.0551	0.0694
term_years	0.0103	0.0053
historical_defaultUnknown	−7.7774	−289.2830
historical_defaultY	−2.9251	−3.9690
cred_hist_length	−0.0834	−0.1276

Table 7: Comparison of Model Performance

Model	ncoef	Rsqa	AUROC	Brier
Non Penalized Full Model	21	0.7307	0.9818	0.2143
LASSO Model	19	0.7110	0.9789	0.2220

4.3 Random forest

Another valuable form of modelling, but in this case non parametric, is Random Forest. This uncorrelated tree generator model is highly reliable due to the uncorrelation property of trees. To perform the algorithm, all variables available in the data set are considered due to the relatively small sensitivity this algorithm has with the covariates included. The main two hyper parameters optimized in this algorithm

are the number of trees and the number of variables considered in the feature boosting. The optimal number of variables are 4 out of the 11 variables available. The optimization decision was based on out-of-bag predictive error. Therefore, 4 variables lead to the best performance of out-of-bag errors. Also, the number of trees was set to 750, since classification errors stopped improving with more than this number of trees.

In terms of variable importance, it is remarkable the importance historical background of costumers have in the possibility of potential default in loans. In fact, this historical default background has a very dominant effect compared to the other control variables considered. See figure 3 to understand visually.

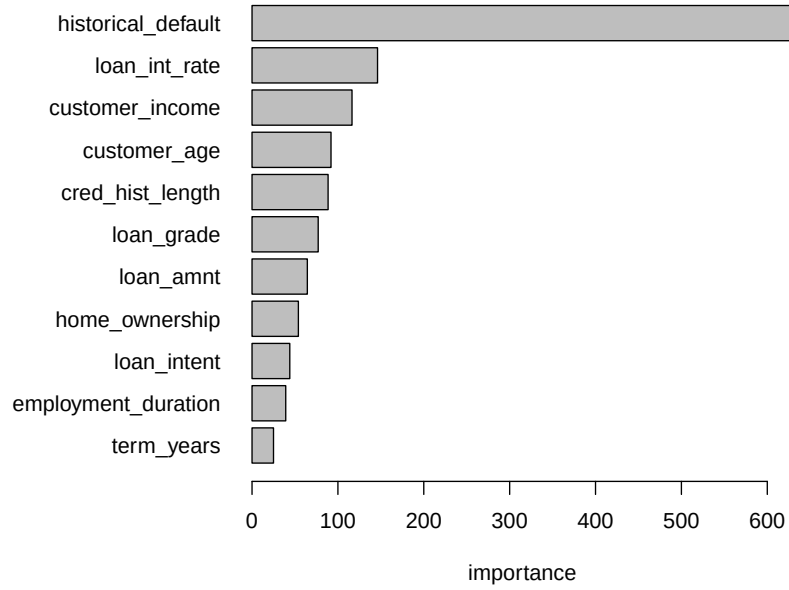


Figure 3: Continuous control variables vs outcome

Table 8: Comparison of Model Performance

Model	ncoef	Rsq	AUROC	Brier
Full Model	21	0.7357	0.9818	0.2143
LASSO Model	19	0.7158	0.9789	0.2220
Random Forest	NA	0.8245	0.9902	0.1749

For the sake of comparison, Table 8 shows AUROC and Brier measures for the three models performed previously. Now, Random Forest performs significantly better compared to the two other parametric models. This due to the AUROC (0.9902) and Brier (0.1749) values, suggesting important improvement in classification prediction and minimizing predictive errors. In fact, Random Forest performs nearly as a perfect model.

4.4 Gradient Boosting

Gradient Boosting is an additional non parametric model for predictions. Is is well know for it's complexity due to the sequential corrections of errors, but also for the well performance in predictions. The approach to selecting the hyper-parameters was as follows: shrinkage was initially set as the minimum possible, 0.01. This implied heavier computational work, although it was still feasible. Due to the relation in shrinkage and number of trees, 1999 trees was the number of trees that best performed CV errors. Therefor, 1999 trees performed in Gradient Boosting correctly addresses the over fitting risk. After this, trials were performed for the decision of Sub sampling Percentage hyper parameter, η . In fact, $\eta = 0.35$ best performed Cross-Validation errors. Other hyperparameters set where maximum of trees to 3, minimum number of observations in node as 10. These are common sets in Gradient Boosting models.

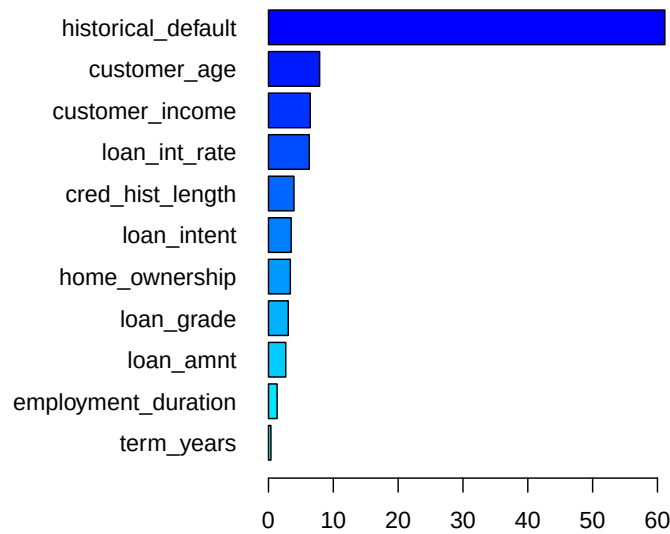


Figure 4: Variable importance: Gradient Boosting

Figure 4 reveals the significant dominance of historical background in default status as the main variable helpful to reduce errors in estimation.

Table 9: Comparison of Model Performance

Model	ncoef	Rsq	AUROC	Brier
Full Model	21	0.7357	0.9818	0.2143
LASSO Model	19	0.7158	0.9789	0.2220
Random Forest	NA	0.8245	0.9902	0.1749
Gradient Boosting	NA	0.8247	0.9905	0.1747

On the same line with reporting measures over classification prediction, Table 9 show measures over model performance, now considering Gradient Boosting. Both algorithmic models perform basically the

same, suggesting that both algorithmic models best performs predictive errors overall the models due to their almost perfect AUROC values and low Brier.

5 Prediction

The objective of the models presented in the previous sections was to estimate the most accurate relationship between the customer’s loan default status and the available covariates, in order to make predictions based on new live data. Nevertheless, these models are estimated in terms of probabilities, but predictions are stated in a binary default or no default status. To perform so, a cutoff value is set for each model performed. The rule chosen for the cutoff decision is the unbiased cutoff value. Table 10 shows the value for each model.

Table 10: Cutoff Values for Classification Prediction

Percentile	Logistic Regression	LASSO Regression	Random Forest	Gradient Boosting
78.92899%	0.4735	0.4633	0.4454	0.4182

After all this said, it is heavily important to understand the performance of each model in predicting classification. This is fairly expressed by the confusion matrix. Table 11 shows measures to evaluate the goodness of the models for classification model prediction.

Table 11: Comparison of Model Accuracy

Model	Accuracy	95% CI	Sensitivity	Specificity
Full Model	0.9441	(0.9368, 0.9508)	0.8674	0.9646
LASSO Model	0.9395	(0.9319, 0.9464)	0.8564	0.9617
Random Forest	0.9995	(0.9983, 0.9999)	0.9984	0.9997
Gradient Boosting	0.9688	(0.9632, 0.9738)	0.9260	0.9802

It is clear that Random Forest is the most reliable model mostly due to the ideal metrics near the “perfect model” for classification prediction. **Therefore, prediction on live data is performed with the Random Forest model, while all models are considered strong alternatives. Specially the algorithmic models.**

6 Conclusion

This analysis aimed to identify significant factors influencing loan defaults and to evaluate the robustness of these factors across different modeling techniques. The key findings from the analysis are as follows:

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- **Significant Variables:** Variables such as customer age, home ownership, employment duration, loan intent, loan grade, loan amount, interest rate, loan term, historical default, and credit history length significantly impact the probability of loan default. Among these, historical default status of customers is the most critical factor in predicting loan defaults, followed by loan grade and home ownership status.
 - **Model Performance:** The Random Forest model outperformed both the logistic regression models (full and penalized LASSO) in terms of overall accuracy and AUROC (Area Under Receiver Operating Characteristic curve) metrics. The Random Forest model demonstrated nearly perfect classification accuracy (99.95%) and the highest AUROC (0.9902), indicating superior predictive performance. In contrast, the full logistic regression model, while still effective, showed slightly lower accuracy (94.41%) and AUROC (0.9818). The Gradient Boosting model also showed strong performance with an accuracy of 96.88% and AUROC of 0.9905, making it a competitive alternative to Random Forest.
 - **Variable Importance:** The importance of the historical default status of customers was evident across all models, highlighting its dominant effect on the likelihood of loan default. Other important factors included loan grade and home ownership status. These findings were consistent across both parametric and non-parametric models.
 - **Improvements and Considerations:** Addressing outliers in variables such as income and employment duration may further enhance model performance.

Overall, the analysis indicates that for this particular dataset, non-parametric models like Random Forest and Gradient Boosting may offer superior predictive accuracy. However, parametric models, particularly logistic regression, also provide robust and interpretable results. Future work should focus on refining model configurations and addressing technical challenges to further improve prediction accuracy and reliability.